SESSION 2 · WEEK 3

Feedback on your proposals

Deliverable 1: team charter and topic proposal

Friday 6 November 2026 · 14:00 CET

Ulysseus EUROPEAN UNIVERSITY

UNIVERSIDAD DE SEVILLA

TECHNICAL UNIVERSITY OF KOŠICE

UCG Univerzitet Crne Gore

AGENDA · 45 MINUTES

What we will do today



10 min	General feedback: what worked and what most teams need to fix.
35 min	Break-out rooms: each team with its team tutor. Your proposal, your data and your questions. When your team has finished, you can leave.

OVERVIEW

A good start

7 of 7
teams submitted their proposal

7
different questions: jobs, energy, housing, growth, pensions…

3
universities working together in every team

What already works

- Most teams have clear roles and a communication channel.
- The topics are relevant for Spain, Slovakia and Montenegro.
- Several teams already spotted the data gaps for Montenegro.

The next slides are the points that most teams need to fix before deliverable 2.

Use indicators that are in the COIL dataset

Not in the dataset

- House prices and housing costs
- Energy prices and energy poverty
- Migration and brain drain
- R&D spending, over-qualification

What we ask

- Check the Data section: 21 indicators, 32 countries, 2010–2025.
- Write the code of every indicator, e.g. SL.UEM.1524.ZS.
- Outside sources are allowed: name the source and upload the data with deliverable 2.

If your key variable is not in the dataset, decide now: change it or find a documented source.

Check Montenegro first

Indicator	Code	Spain	Slovakia	Montenegro
Education spending (% GDP)	SE.XPD.TOTL.GD.ZS	2010–2022	2010–2022	no data
Tax revenue (% GDP)	GC.TAX.TOTL.GD.ZS	2010–2024	2010–2024	no data
CO2 emissions per capita	EN.GHG.CO2.PC.CE.AR5	2010–2024	2010–2024	no data
Renewable energy (%)	EG.FEC.RNEW.ZS	2010–2021	2010–2021	2010–2021
Gini index	SI.POV.GINI	2010–2023	2010–2023	2010–2021

All other indicators cover the three countries from 2010 to at least 2023.

A question your data can answer

Too broad

"How do education and technology shape the future of European economies?"

Answerable

"Is internet use associated with GDP per capita across 32 European countries, 2010–2024?"

Two simple rules

- Every concept in your question needs a variable in your data.
- Write "is associated with", not "impacts" or "causes": 15 years per country show associations, not effects.

4 · SAMPLE

Use the whole of Europe, not only three countries

$3 \times 15 = 45$

observations with Spain, Slovakia and Montenegro only

$32 \times 15 = 480$

observations with all European countries

Why it matters for deliverable 2

- A regression or a clustering with 3 countries says very little.
- Estimate with all countries, then show where Spain, Slovakia and Montenegro fall.
- Deliverable 2 asks for this comparison with the rest of Europe.

Make your rules concrete

Fixed meeting

One weekly day and time (CET), not “to be agreed” or a single date.

Answer time

A number: 24 or 48 hours, also at weekends.

If someone does not contribute

Talk inside the team first, agree a new deadline, and only then tell your tutor. Never “exclude the member”.

Update your charter now: you will need it in the coming weeks.

6 · FORM

The AI note and the template

AI note

- Say what you did, not what you might do.
- Which tool, for what task, and how you checked the result.
- AI is not a data source: every number needs a real source.

Template

- Delete the instructions and every [bracket].
- Keep only your part: 2 pages at most.
- End the research question with a question mark.
- File name: Team X – Deliverable 2.pdf

FOR YOUR WEEK-4 TUTORIAL

Two decisions to bring to your tutor

1. Which software?

- One tool for the whole team: Python, R, Gretl or Power BI.
- Who has it installed and has opened the COIL file?
- Regression in Power BI needs extra steps: plan for it.

2. Who does what?

- A named person for each task until deliverable 2:
- data and dictionary · descriptive charts · regression · PCA and clustering · report.
- Mix universities in every task.

Roles are a start; a task list with names and dates is what keeps the team on track.

ONE MORE THING

Wanted: three team names

TEAM 1	TEAM 2	TEAM 3	TEAM 4	TEAM 5	TEAM 6	TEAM 7
?	Global Analyzers	Analytical Staff	Brainstorm	?	I Migliori	?

Teams 1, 5 and 7: you deserve better than a number.

Choose a name and send it to Daniel (druiz8@us.es). It will appear on the website.

Bonus points for creativity. Not real points, sadly.

YOUR NEXT STEP · WEEK 4

Start with descriptive statistics

Before any regression: look at your data. Bring this to your week-4 tutorial.

1 **Summary table**

Mean, median, min, max and n of every variable. Report missing values.

2 **Spain, Slovakia, Montenegro**

The same table for each country, next to the European average.

3 **Three charts**

Over time for the three countries; a scatter plot of your two main variables; one comparison with the rest of Europe.

4 **One sentence each**

What does each table and chart show? Write it under it, in plain English.

How to do it in Gretl, Power BI, Python or R: data workshop slides (Sessions section of the website).

NOW AND NEXT

Break-out rooms and next steps

Team tutor	Teams
Alena Mojsejová	1, 2, 3
Daniel Ruiz Romera	4, 6
Bojan Pejović	5, 7

Next steps

- Week 4: descriptive statistics, and book a short tutorial with your team tutor.
- Fix your indicators and question before you analyse.
- Deliverable 2: Sunday 29 November, 23:59 CET.

Bring your questions to the break-out room. When your team has finished with your tutor, you can leave.

BREAK-OUT ROOMS ARE OPEN

Join the room of your team

Team 1 Tutor: Alena	**Team 2 · Global Analyzers** Tutor: Alena	**Team 3 · Analytical Staff** Tutor: Alena	**Team 4 · Brainstorm** Tutor: Daniel
Team 5 Tutor: Bojan	**Team 6 · I Migliori** Tutor: Daniel	**Team 7** Tutor: Bojan	

In Teams: Rooms > your team > Join room. Cannot get in? Write your team number in the chat and Daniel will move you.

When your team has finished with your tutor, you can leave. Thank you!